LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 12. Sets

Lecture: 126.Sets - Introduction

Section 1: Lecture Summary

This lecture introduces the Python data type **set**, covering its definition, properties, methods of creation, and key behaviors. The instructor explains that a set is a **collection of distinct elements** enclosed in **curly braces**, emphasizing that sets are **unordered**, **heterogeneous**, and **mutable**. Examples illustrate creation using curly braces and the `set()` function with iterables such as lists and strings. Important constraints like the inability to store mutable elements (e.g., lists) inside a set are highlighted. The lecture also clarifies that empty curly braces represent an empty dictionary, not a set, requiring at least one element or the use of `set()` for an empty set. The unordered nature means sets do not support indexing or slicing, and printing may misleadingly display elements in sorted order, but iteration via loops reveals their actual arbitrary order. Methods for modifying sets, such as `add()`, `remove()`, and `pop()`, are demonstrated with examples. The lecture concludes by teasing a follow-up session explaining internal set representation.

Section 2: Key Concepts and Explanations

- **Set Definition**

  - A set is a collection of **distinct elements** enclosed within **curly braces `{}`**.

  - Example: `{1, 2, 3, 4}`.

- **Properties of Sets**

  - **Distinct elements**: Duplicate values are automatically removed.

  - **Unordered**: Sets do not maintain any insertion order and do not allow indexing or slicing. The elements are stored and represented in an unpredictable order.

- **Heterogeneous**: Sets can contain elements of different data types (integers, floats, strings, booleans, tuples).

- **Mutable**: Sets can be modified after creation by adding or removing elements, but the elements themselves must be immutable.

- **Creation of Sets**

  - Using curly braces with one or more elements: `S = {1, 2, 3}`

  - Using the `set()` constructor with an iterable argument such as list or string:

    - `set([1, 2, 3])` creates `{1, 2, 3}`

    - `set('python')` creates a set of distinct characters like `{'p', 'y', 't', 'h', 'o', 'n'}`

  - **Important**: Empty curly braces `{}` creates an empty dictionary, not a set. To create an empty set, use `set()`.

- **Mutable vs Immutable Elements in Sets**

  - Elements stored inside a set must be **hashable** (immutable).

  - Allowed: integers, floats, strings, tuples (immutable)

  - Not allowed: lists or other mutable types (cause `TypeError: unhashable type: 'list'`)

- **Unordered Nature and Display**

  - Sets do not have indices, so you cannot access elements by position or slice them.

  - Printing a set in Jupyter Notebook often shows elements in sorted order, but the actual stored order is arbitrary as seen when iterating using a loop.

  - Adding elements does not guarantee any particular storage order.

- **Set Methods Demonstrated**

  - `add(element)`: Adds an element to the set. Where it is stored internally is unpredictable.

  - `remove(element)`: Removes an element; raises error if the element is not found.

- `pop()`: Removes and returns an arbitrary element from the set (not necessarily the last one added since no order is maintained).

Section 3: Example Code and Use Cases

Creating sets with curly braces and set() function

```
S1 = {1, 2, 3, 4, 5, 6}
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = {5}

print(type(S1))  # <class 'set'>
print(S3)        # e.g. {'h', 'n', 'o', 'p', 't', 'y'}
```

Creates sets using curly braces and `set()` from a list and a string. Demonstrates distinct elements from string characters.

Empty set creation - correct and incorrect

```
empty_set_wrong = {}    # This creates an empty dictionary
print(type(empty_set_wrong))  # <class 'dict'>

empty_set = set()       # Correct way to create an empty set
print(type(empty_set))  # <class 'set'>
```

Shows that empty curly braces create a dictionary, empty set must be created with `set()`.

Demonstrate set unordered nature by printing elements in a loop

```
S = {10, 20, 30, 40, 50}
for elem in S:
    print(elem)
```

Output order is arbitrary and may differ from insertion order.

Adding elements to a set (mutable)

```
S = {10, 20, 30, 40, 50}
S.add(60)          # Add integer
S.add('hi')        # Add string
```

```
S.add((1, 2, 3))   # Add tuple

print(S)
```

Demonstrates adding various immutable types to a set.

Trying to add a mutable element (list) throws an error

```
S = {10, 20, 30}
try:
    S.add([1, 2, 3])
except TypeError as e:
    print(e)  # Output: unhashable type: 'list'
```

Shows that mutable elements cannot be added to a set.

Removing and popping elements from a set

```
S = {10, 20, 30, 40, 50, 60}
S.remove(20)   # Removes 20
print(S)

popped = S.pop()  # Removes an arbitrary element, not predictable which
print("Popped element:", popped)
print(S)
```

Demonstrates removing specific elements and popping arbitrary items from sets.

Section 4: Key Takeaways

- A **set** is a **collection of distinct, unordered, heterogeneous elements** enclosed in curly braces or created with the `set()` function.

- Sets **do not support indexing or slicing** because they are unordered and have no positional indices.

- **Empty curly braces create a dictionary**, not a set; to create an empty set, use `set()`.

- Sets are **mutable**, meaning you can add or remove elements after creation.

- Elements in a set must be **immutable (hashable)**; mutable types like lists cannot be elements of a set.

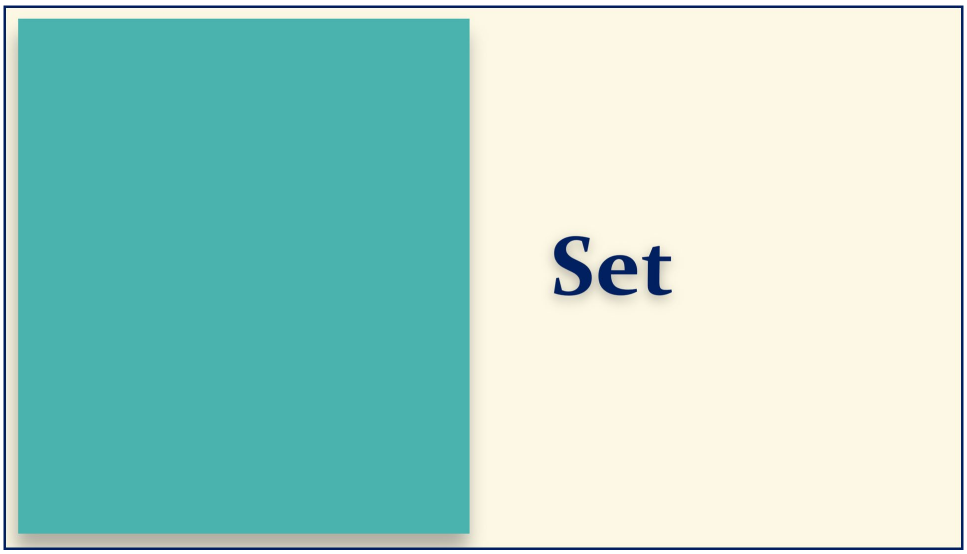
- Printing a set may show elements in sorted order, but the actual internal order is arbitrary; looping reveals true iteration order.

- Use `add()`, `remove()`, and `pop()` methods to modify sets.

- Understanding the concept of hashing and how sets store data internally will be covered in the next lecture.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Set Introduction

Set Introduction

Set Introduction

What is a Set?

Set Introduction

What is a Set?

Creation

Set Introduction

What is a Set?

Creation

Unordered

Set Introduction

What is a Set?

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What is a Set?

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Set Introduction

L1 = [1, 2, 3, 4, 5, 6]

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

L1 = [1, 2, 3, 4, 5, 6]

T1 = (1, 2, 3, 4, 5, 6)

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

L1 = [1, 2, 3, 4, 5, 6]

T1 = (1, 2, 3, 4, 5, 6)

S1 = { 1, 2, 3, 4, 5, 6 }

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

$S1 = \{ 1, 2, 3, 4, 5, 6 \}$

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

$S1 = \{ 1, 2, 3, 4, 5, 6 \}$

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

S1 = { 1, 2, 3, 4, 5, 6 }

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

Collection of Elements

S2 = { 1, 1, 2, 2, 3, 4, 5 }

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

Collection of Elements

S2 = { 1, ~~1~~, 2, ~~2~~, 3, 4, 5 }

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

Collection of Elements

Distinct Elements

$S2 = \{ 1, \cancel{1}, 2, \cancel{2}, 3, 4, 5 \}$

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

Collection of Elements

Distinct Elements

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

Distinct Elements

S3 = { 1, 2.5, 'Jack', True }

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

Distinct Elements

Heterogenous

S3 = { 1, 2.5, 'Jack', True }

Set Introduction

Set Introduction

What is a Set?

Creation

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Collection of Elements

Distinct Elements

Heterogenous

Set Introduction

What is a Set?

Creation

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Mutable

Set Introduction

Collection of Elements

Distinct Elements

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Indexing S1[2]

Set Introduction

What is a Set?

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Mutable

Set Introduction

Collection of Elements

Distinct Elements

Heterogenous

~~Indexing~~ S1[2]

Set Introduction

Set Introduction

What is a Set?

Creation

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Mutable

Collection of Elements

Distinct Elements

Heterogenous

~~Indexing~~ S1[2]

Slicing S1[1 : 4 : 2]

Set Introduction

Set Introduction

What is a Set?

Creation

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Mutable

Collection of Elements

Distinct Elements

Heterogenous

~~Indexing~~ S1[2]

~~Slicing~~ S1[1 : 4 : 2]

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

Distinct Elements

Heterogenous

Unordered

~~Indexing~~ S1[2]

~~Slicing~~ S1[1 : 4 : 2]

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Collection of Elements

S1 = { 1, 2, 3, 4, 5, 6 }

Distinct Elements

S2 = { 1, ~~1~~, 2, ~~2~~, 3, 4, 5 }

Heterogenous

S3 = { 1, 2.5, 'Jack', True }

Unordered

~~Indexing~~ S1[2]

~~Slicing~~ S1[1 : 4 : 2]

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

- Collection of Elements
- Distinct Elements
- Heterogenous
- Unordered

Set Introduction

What is a Set?

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What is a Set?

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- Distinct Elements
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Its Heterogenous

Set Introduction

What is a Set?

Creation

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Set Introduction

- Collection of Elements
- Distinct Elements
- Heterogenous
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Unordered Collection of Distinct Elements

Its Heterogenous

Its Mutable

Set Introduction

Set Introduction

What is a Set?

Creation

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Mutable

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

S1 = { 1, 2, 3, 4, 5, 6 }

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set(iterable)

Set Introduction

Set Introduction

What is a Set?

Creation

Unordered

Mutable

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set([1, 2, 3, 4, 5, 6])

Set Introduction

What is a Set?

Creation

Unordered

Mutable

Set Introduction

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set([1, 2, 3, 4, 5, 6])

S3 = set('python')

Set Introduction

What is a Set?

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Set Introduction

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set([1, 2, 3, 4, 5, 6])

S3 = set('python')

S4 = set( )

Set Introduction

What is a Set?

Creation

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Set Introduction

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set([1, 2, 3, 4, 5, 6])

S3 = set('python')

S4 = { }

Set Introduction

What is a Set?

Creation

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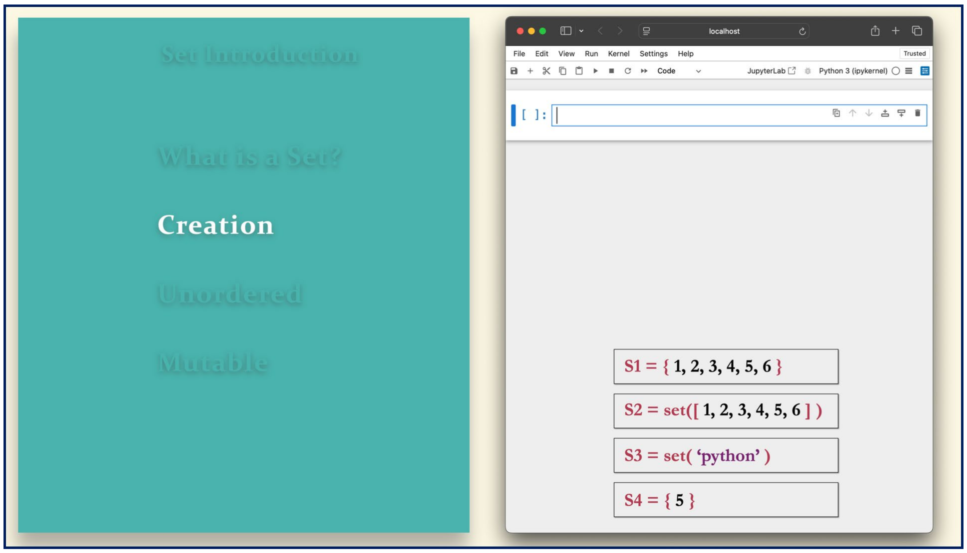
Set Introduction

S1 = { 1, 2, 3, 4, 5, 6 }

S2 = set([1, 2, 3, 4, 5, 6])

S3 = set('python')

S4 = { 5 }



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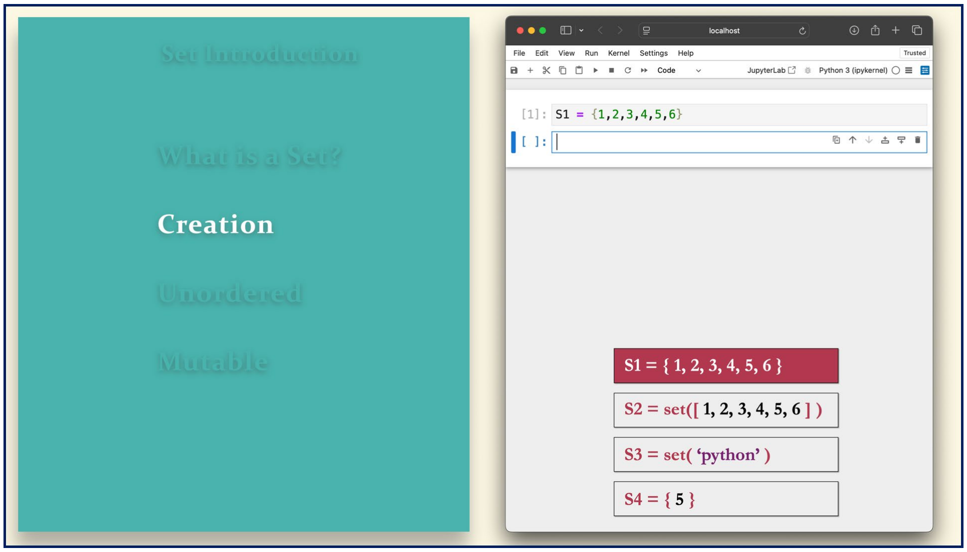
Python 3 (ipykernel)
[ ]:
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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Python 3 (ipykernel)
[1]: S1 = {1,2,3,4,5,6}
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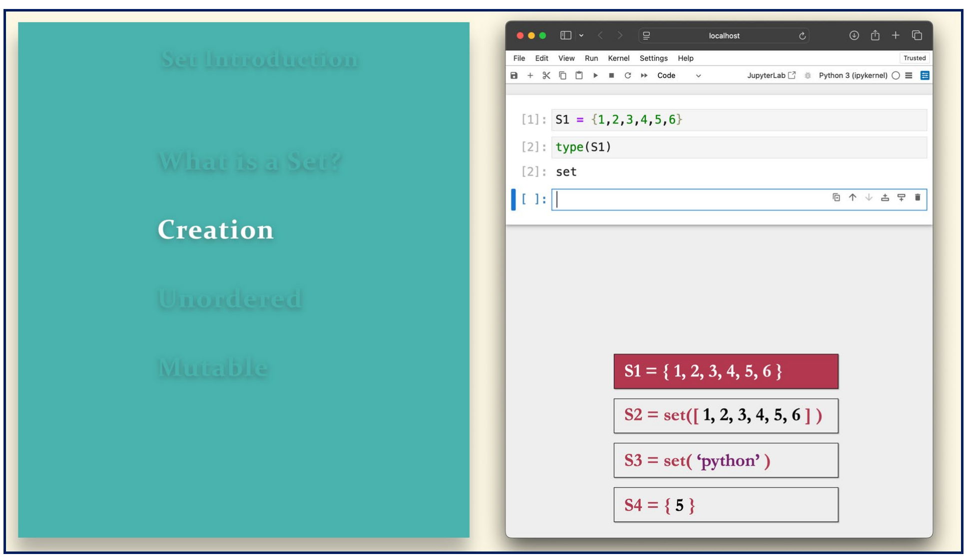
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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Python 3 (ipykernel)
[1]: S1 = {1,2,3,4,5,6}
[2]: type(S1)
[2]: set
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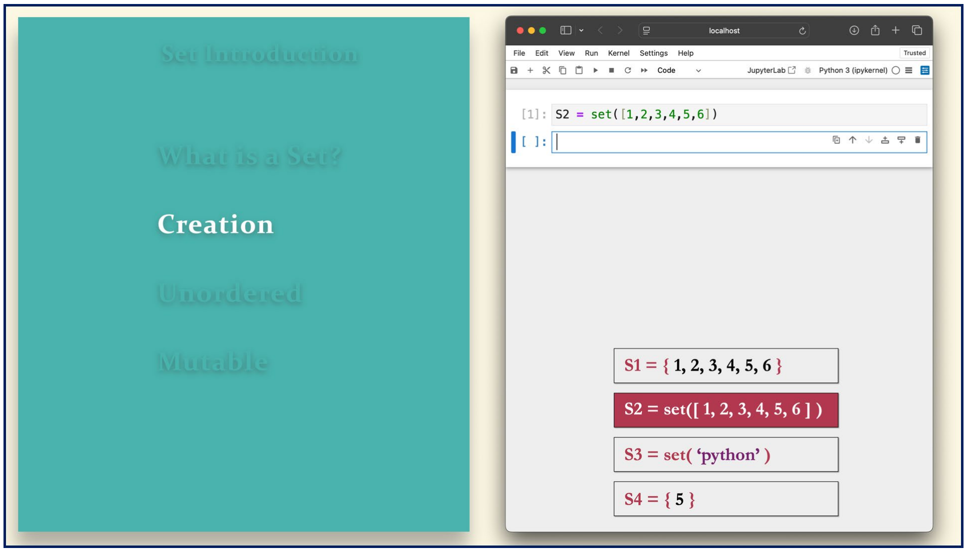
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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[1]: S2 = set([1,2,3,4,5,6])
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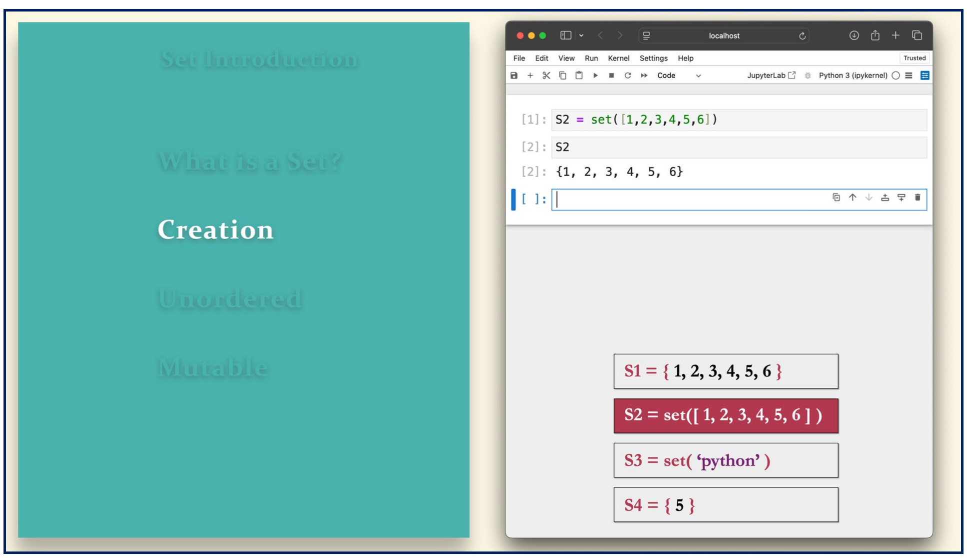
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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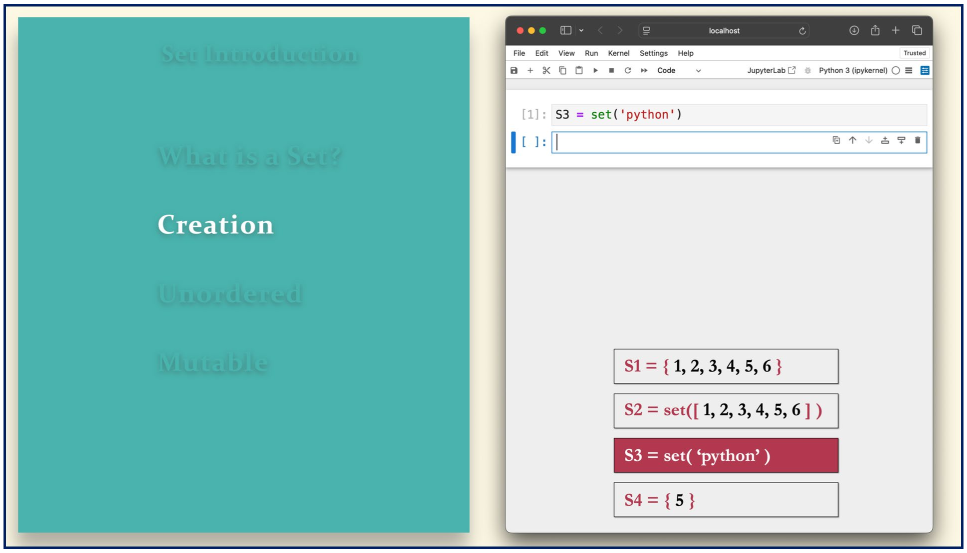
Python 3 (ipykernel)
[1]: S2 = set([1,2,3,4,5,6])
[2]: S2
[2]: {1, 2, 3, 4, 5, 6}
[ ]:
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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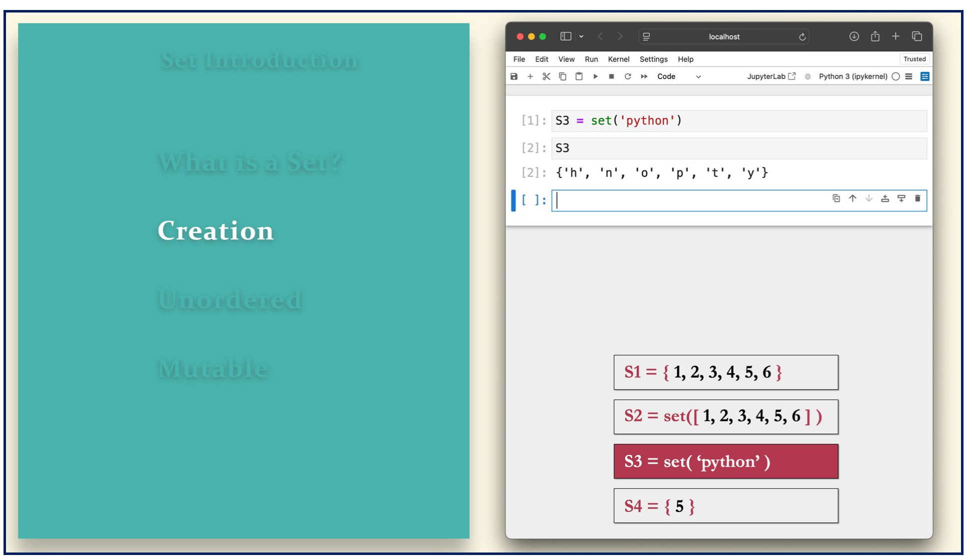
Python 3 (ipykernel)
[1]: S3 = set('python')
[ ]:
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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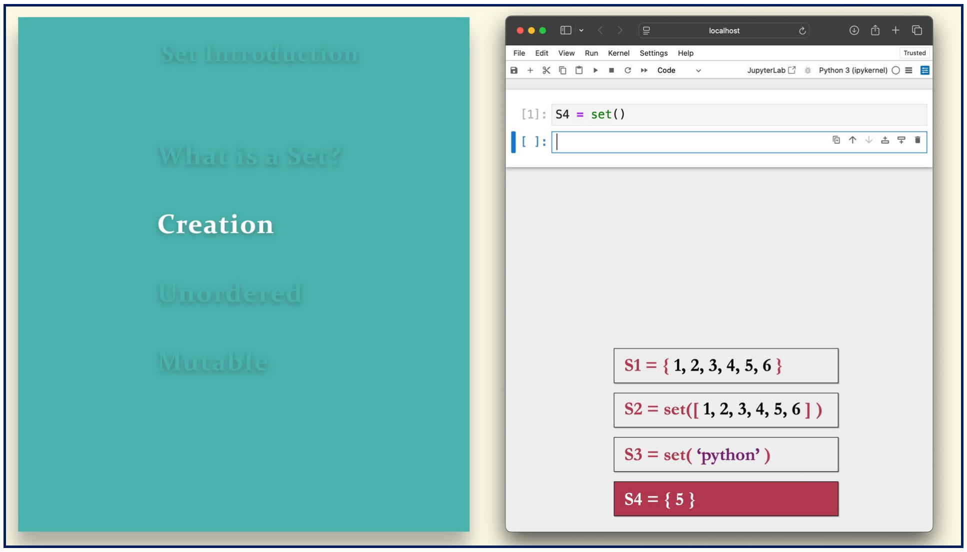
Python 3 (ipykernel)
[1]: S3 = set('python')
[2]: S3
[2]: {'h', 'n', 'o', 'p', 't', 'y'}
[ ]:
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set(‘python’)
S4 = { 5 }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Python 3 (ipykernel)
[1]: S4 = set()
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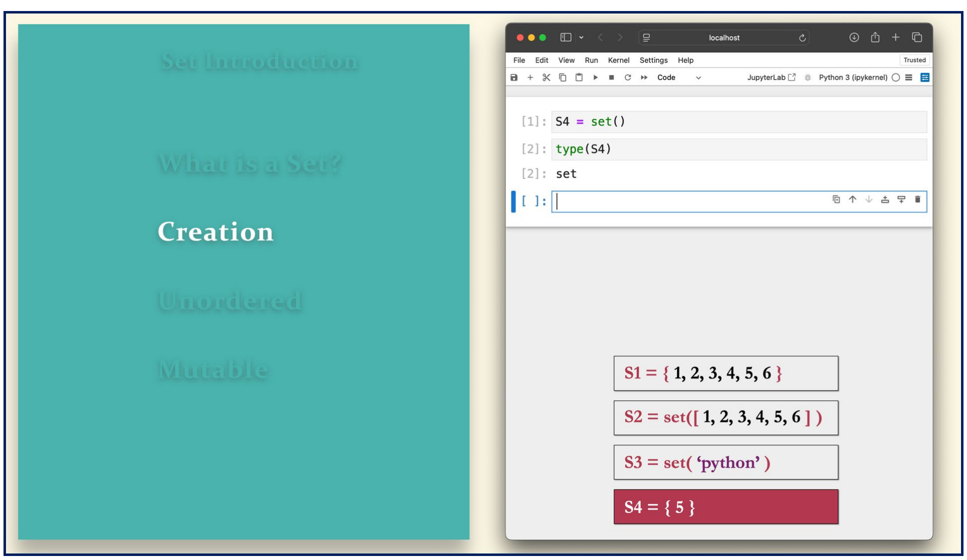
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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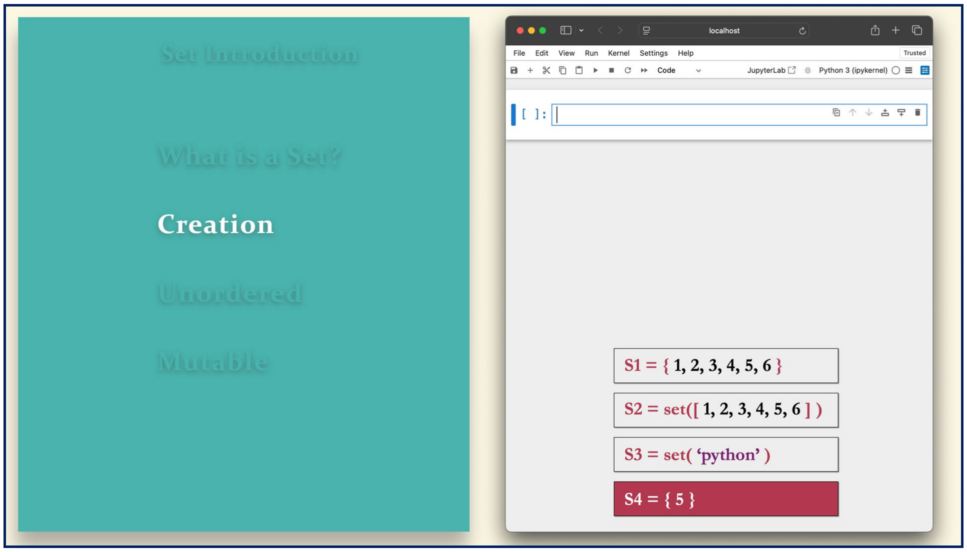
Python 3 (ipykernel)
[1]: S4 = set()
[2]: type(S4)
[2]: set
[ ]:
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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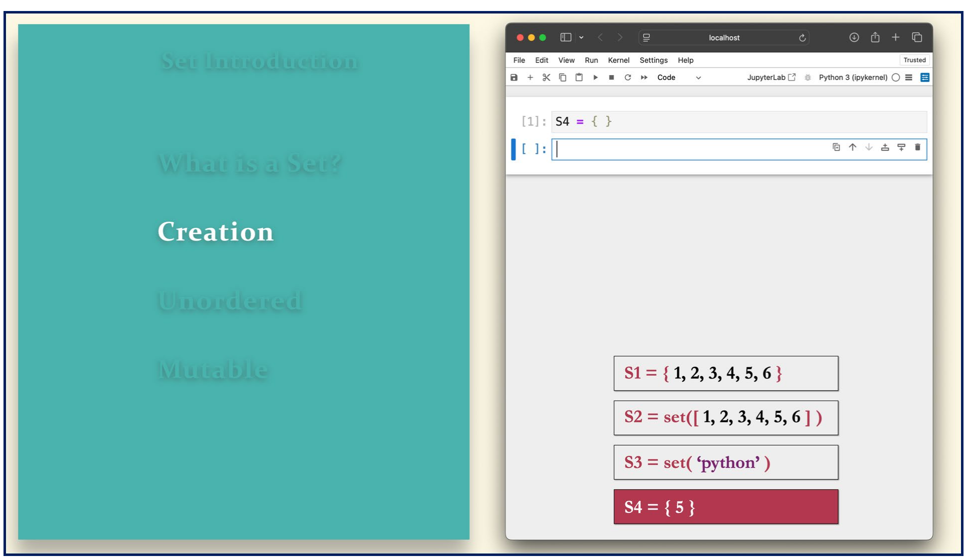
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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Python 3 (ipykernel)
[1]: S4 = { }
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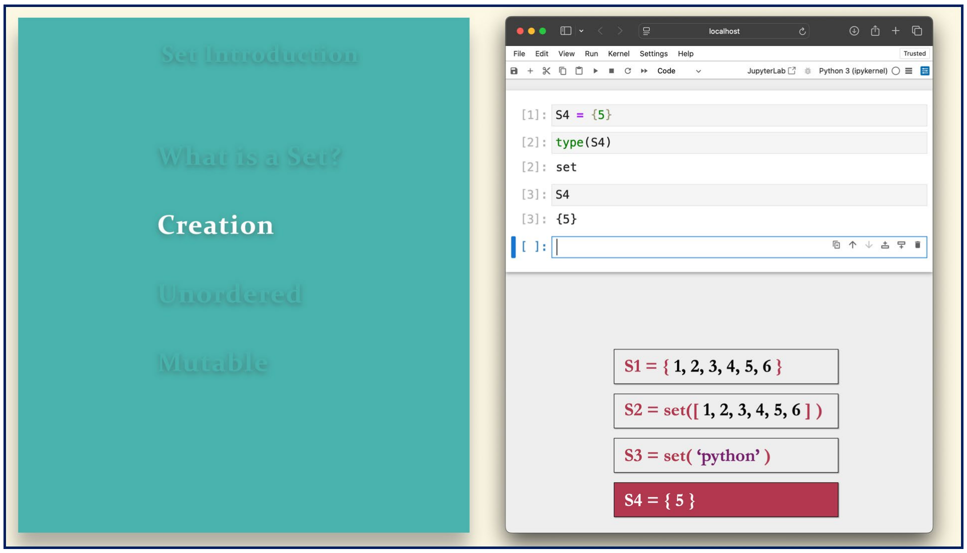
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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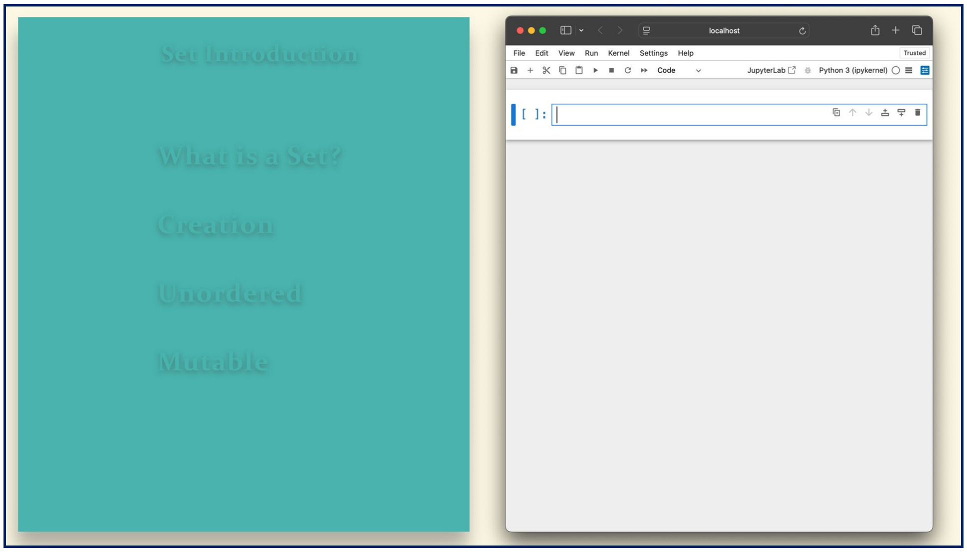
[1]: S4 = {5}
[2]: type(S4)
[2]: set
[3]: S4
[3]: {5}
S1 = { 1, 2, 3, 4, 5, 6 }
S2 = set([1, 2, 3, 4, 5, 6])
S3 = set('python')
S4 = { 5 }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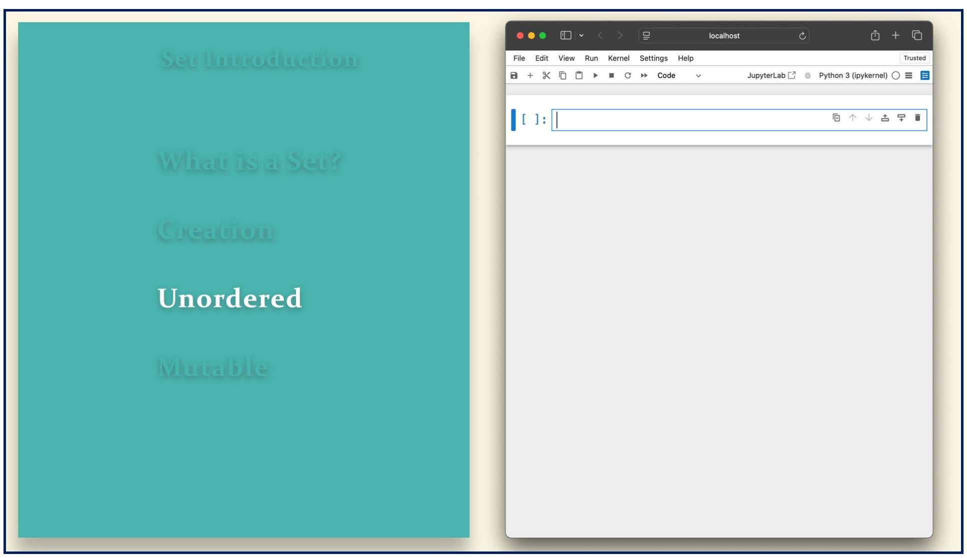
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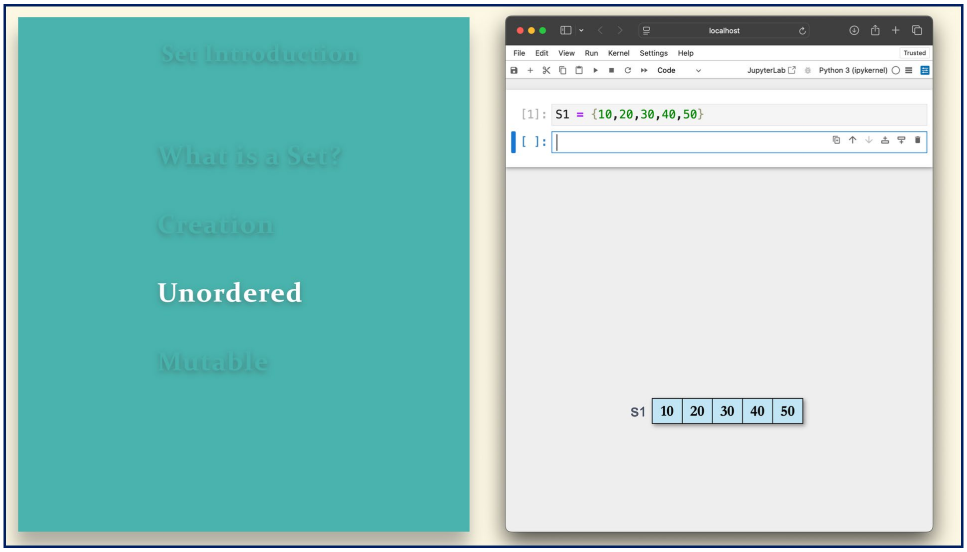
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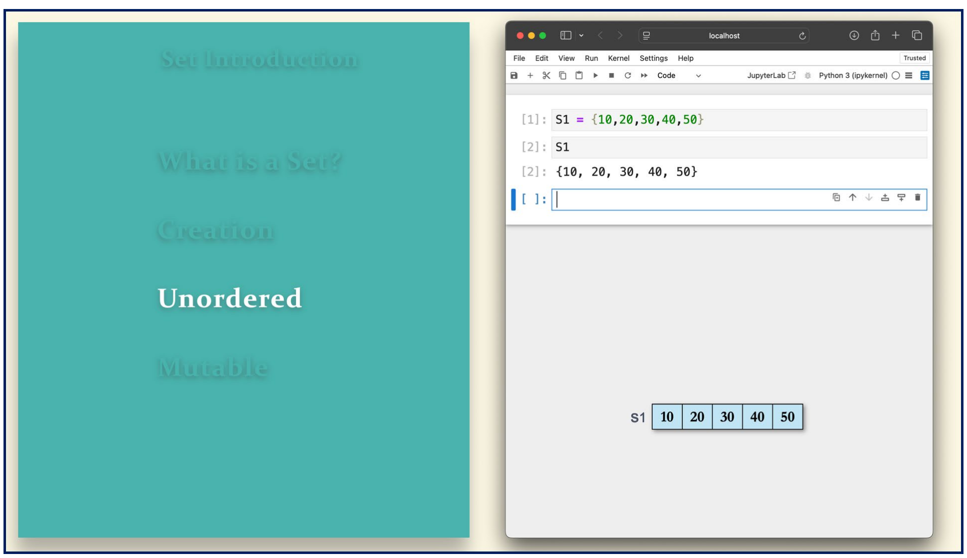
Python 3 (ipykernel)
[1]: S1 = {10,20,30,40,50}
[ ]:
S1
10
20
30
40
50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Python 3 (ipykernel)
[1]: S1 = {10,20,30,40,50}
[2]: S1
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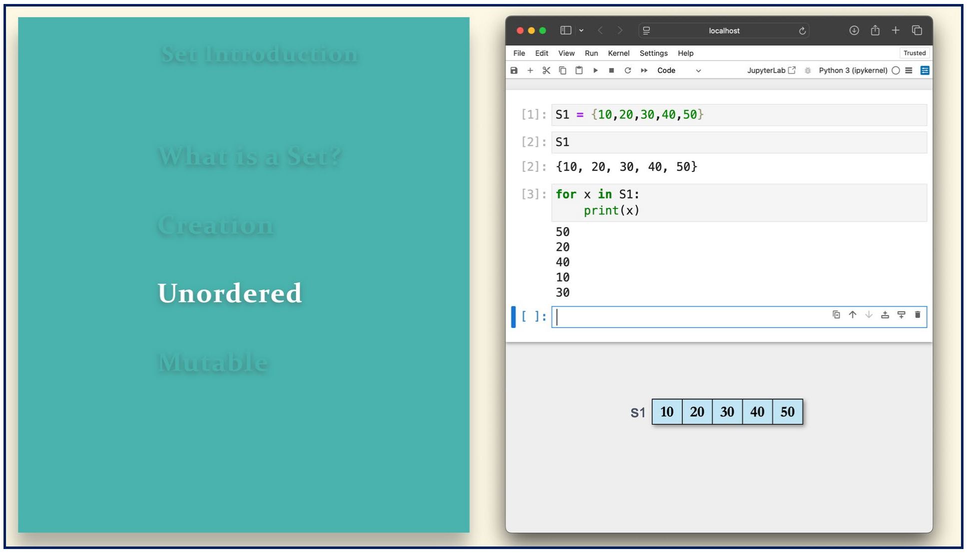
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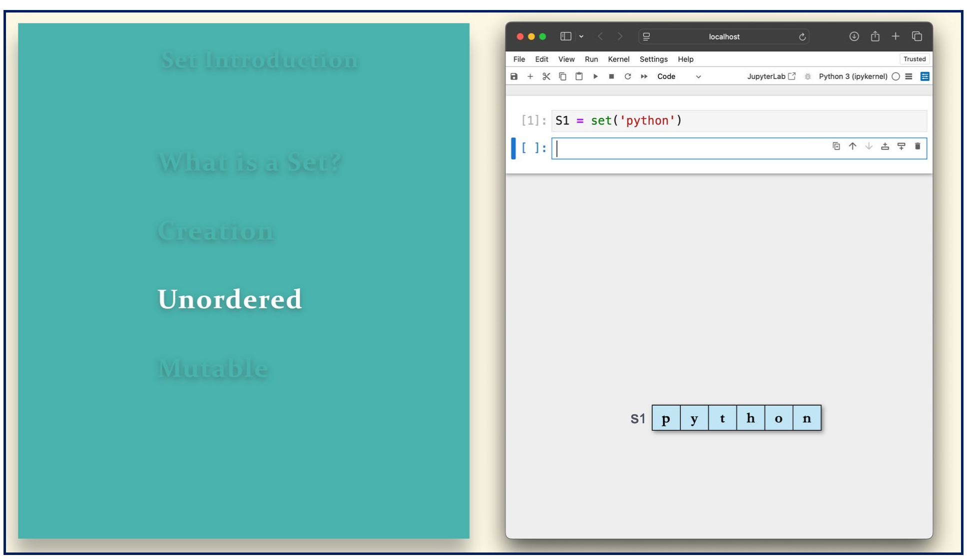
Code
JupyterLab
Python 3 (ipykernel)
[1]: S1 = {10,20,30,40,50}
[2]: S1
[2]: {10, 20, 30, 40, 50}
[3]: for x in S1:
    print(x)
50
20
40
10
30
[ ]:
S1
10 20 30 40 50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Python 3 (ipykernel)
[1]: S1 = set('python')
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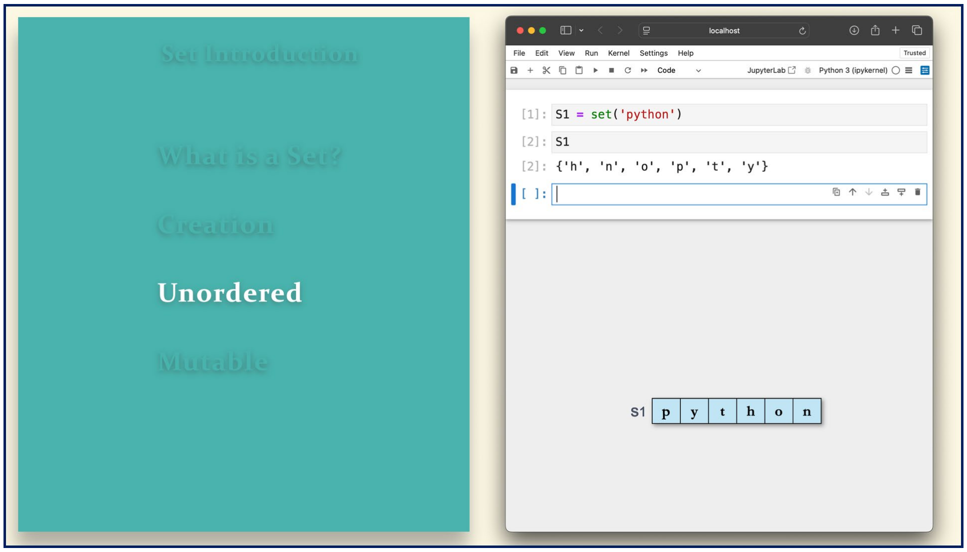
S1
p y t h o n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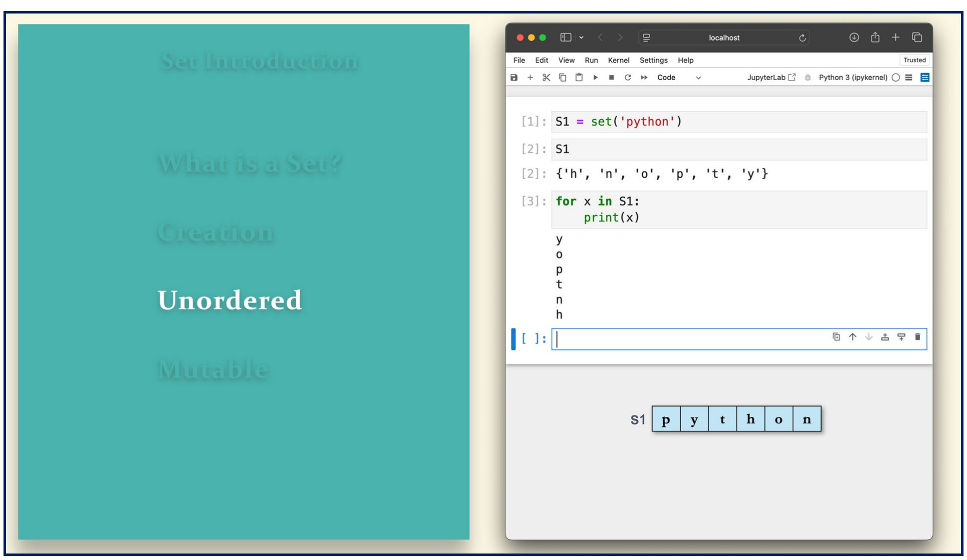
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[1]: S1 = set('python')
[2]: S1
[2]: {'h', 'n', 'o', 'p', 't', 'y'}
[ ]:
S1 p y t h o n


Set Introduction
What is a Set?
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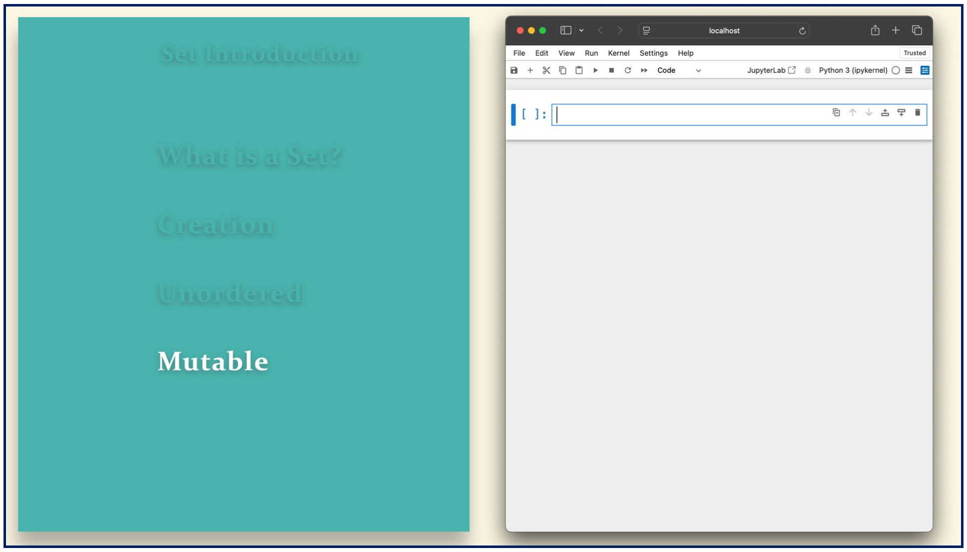
Python 3 (ipykernel)
[1]: S1 = set('python')
[2]: S1
[2]: {'h', 'n', 'o', 'p', 't', 'y'}
[3]: for x in S1:
    print(x)
y
o
p
t
n
h
[ ]:
S1 p y t h o n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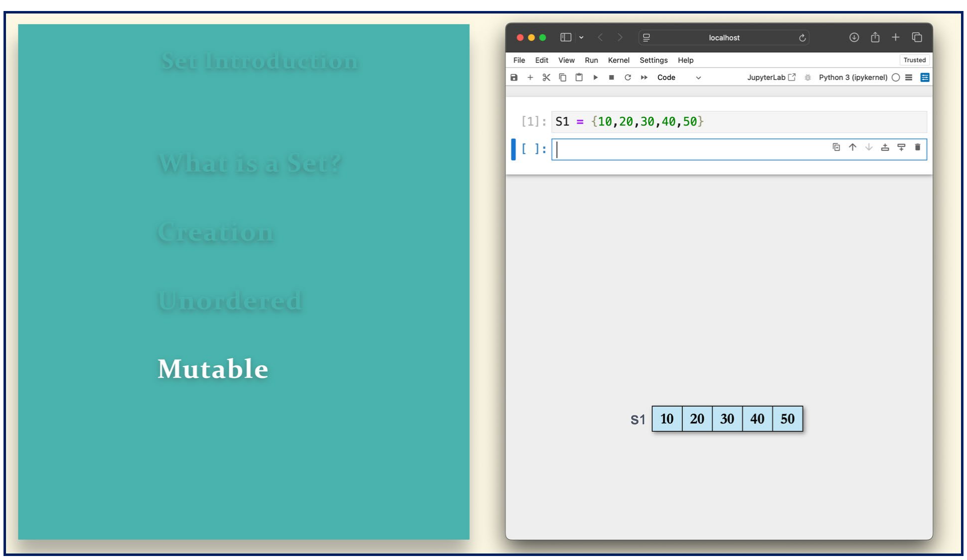
[ ]: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Code
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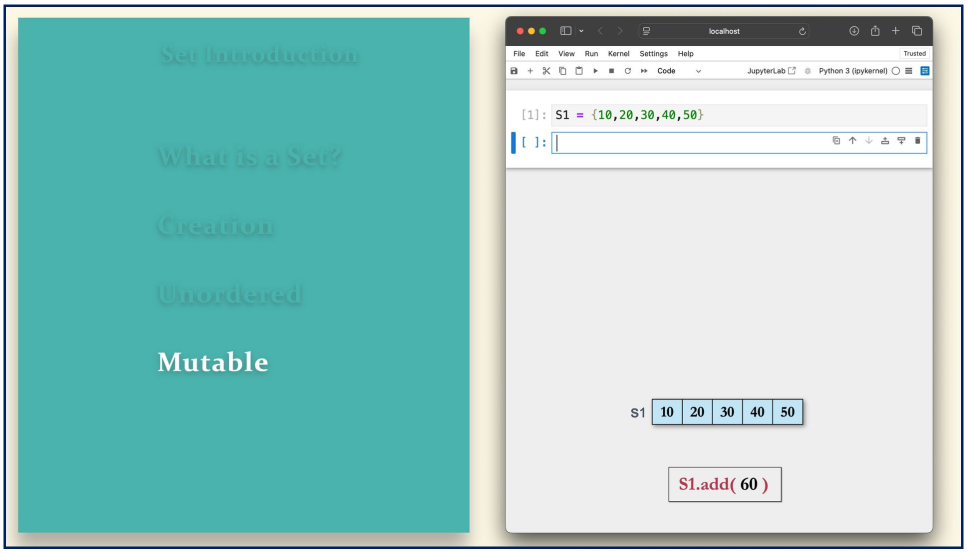
Python 3 (ipykernel)
[1]: S1 = {10,20,30,40,50}
[ ]:
S1
10
20
30
40
50



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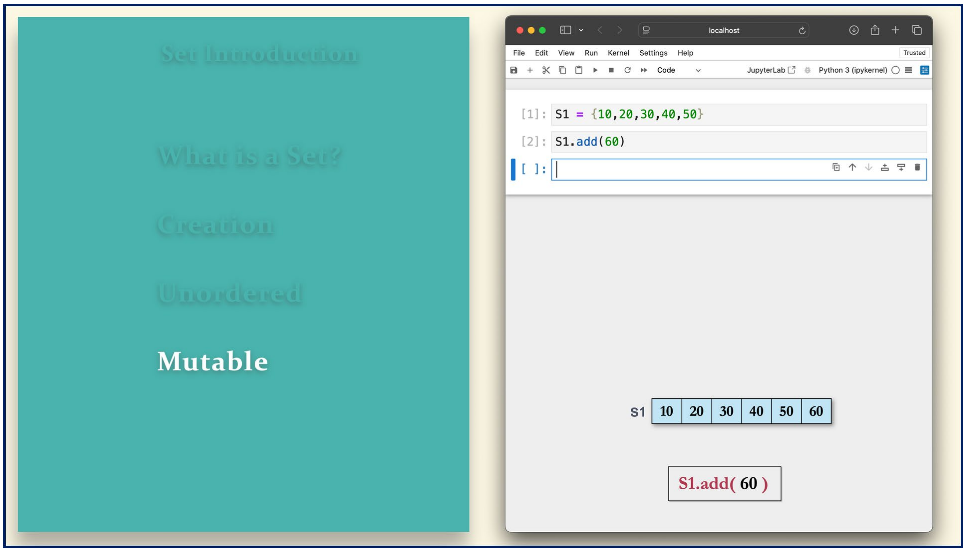
[1]: S1 = {10,20,30,40,50}
[ ]:
S1
10
20
30
40
50
S1.add(60)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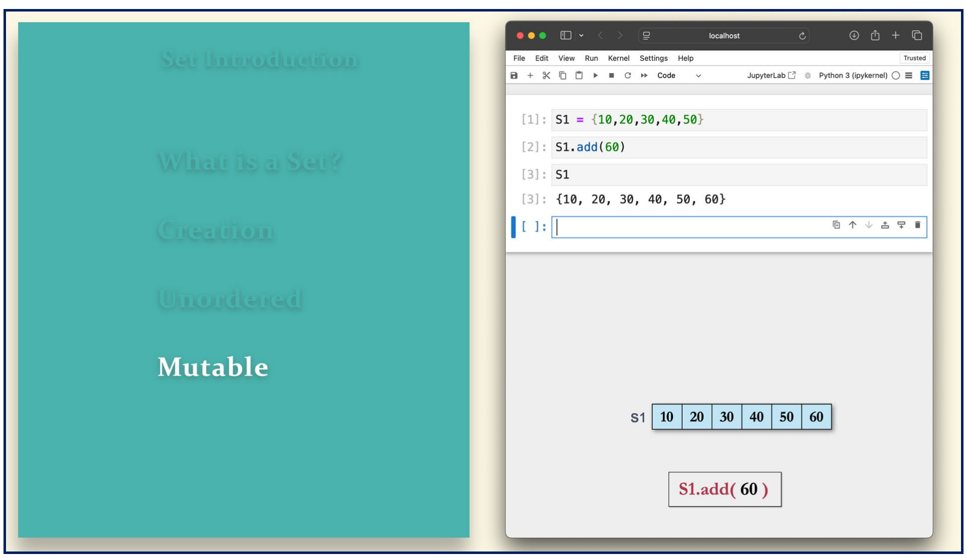
[1]: S1 = {10,20,30,40,50}
[2]: S1.add(60)
[ ]:
S1
10
20
30
40
50
60
S1.add(60)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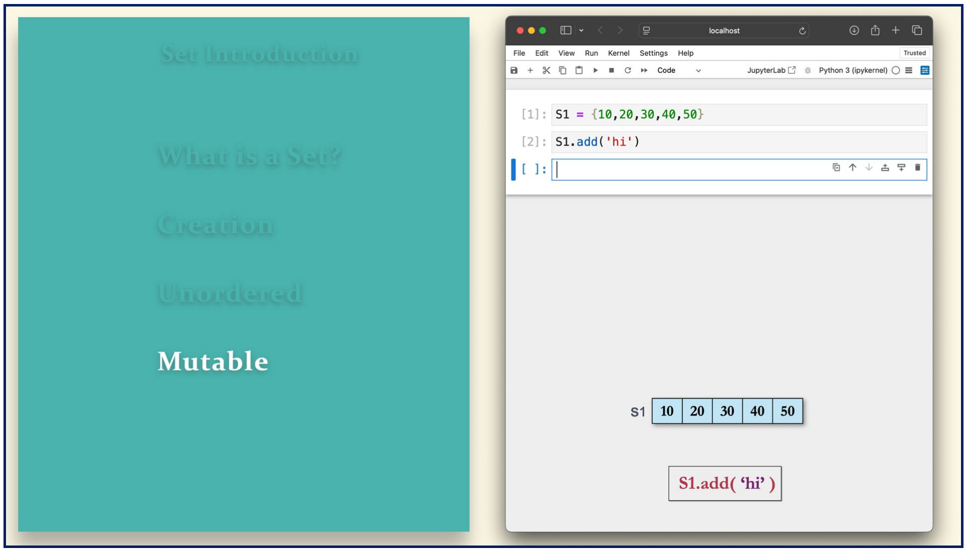
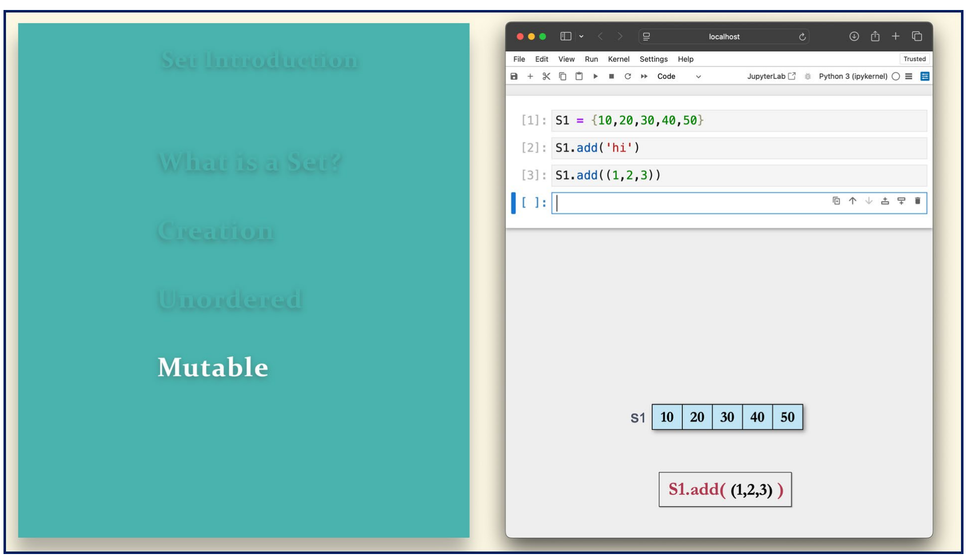
[2]: S1.add(60)
[3]: S1
[3]: {10, 20, 30, 40, 50, 60}
[ ]:
S1
10
20
30
40
50
60
S1.add(60)


Set Introduction
What is a Set?
Creation
Unordered
Mutable
[1]: S1 = {10,20,30,40,50}
[2]: S1.add('hi')
S1
10 20 30 40 50
S1.add('hi')


Set Introduction
What is a Set?
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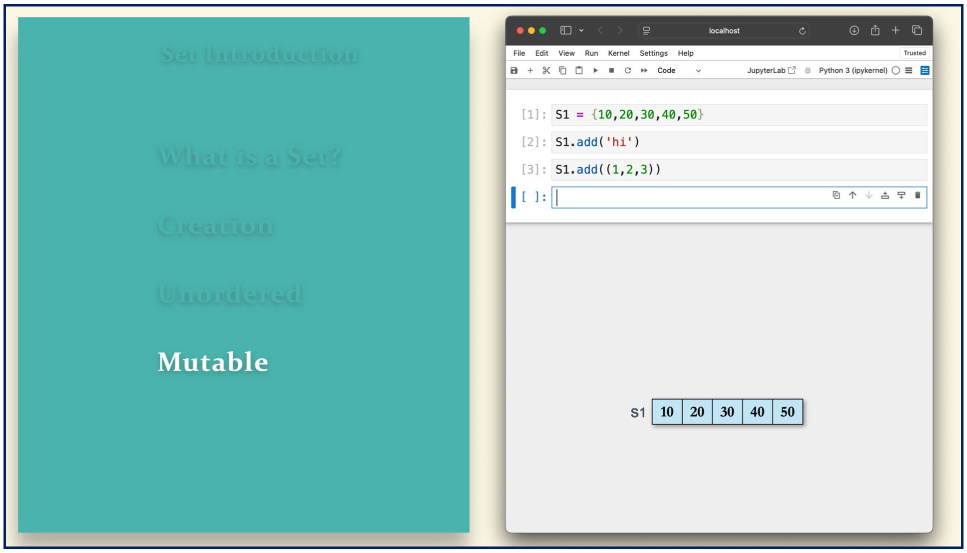
Mutable
[1]: S1 = {10,20,30,40,50}
[2]: S1.add('hi')
[3]: S1.add((1,2,3))
S1
10 20 30 40 50
S1.add((1,2,3))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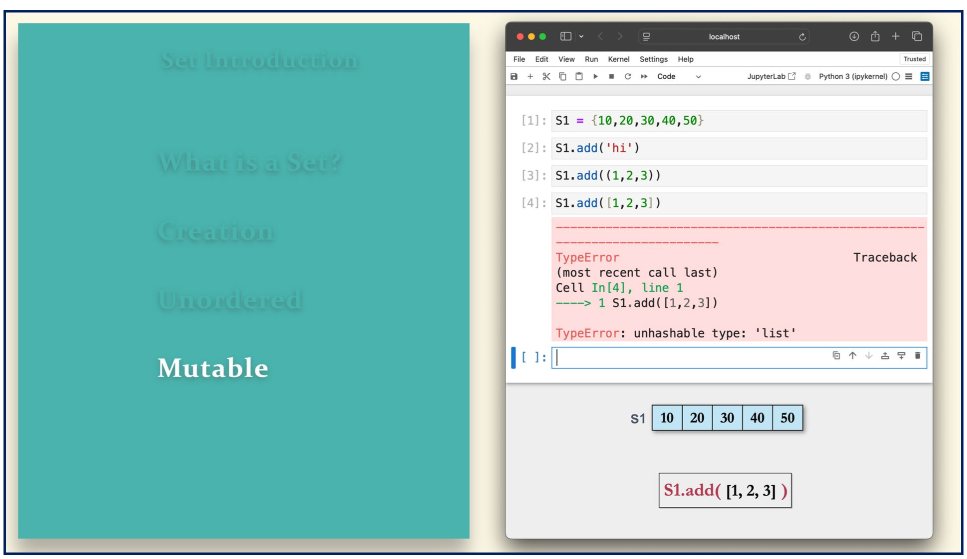
S1 10 20 30 40 50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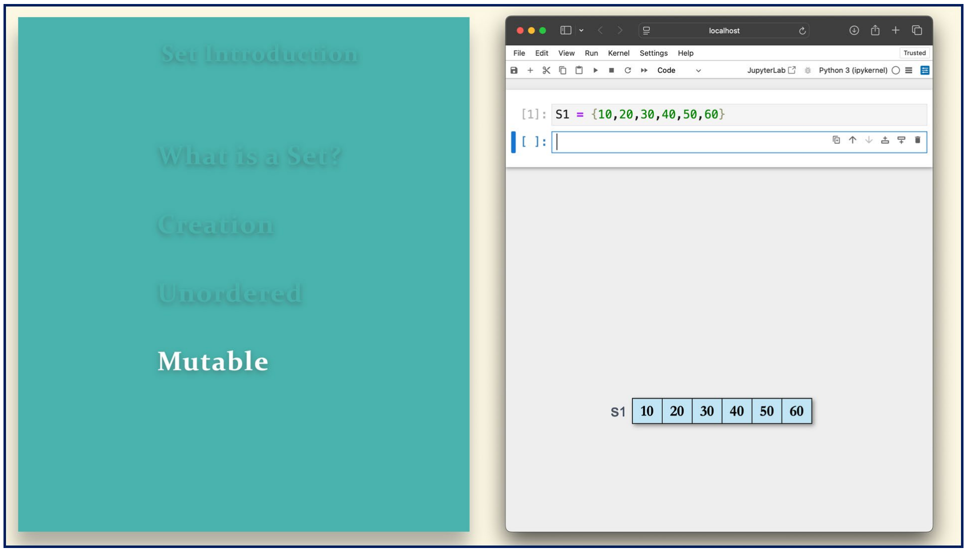
[1]: S1 = {10,20,30,40,50}
[2]: S1.add('hi')
[3]: S1.add((1,2,3))
[4]: S1.add([1,2,3])
TypeError Traceback (most recent call last)
Cell In[4], line 1
----> 1 S1.add([1,2,3])
TypeError: unhashable type: 'list'
S1 10 20 30 40 50
S1.add([1, 2, 3])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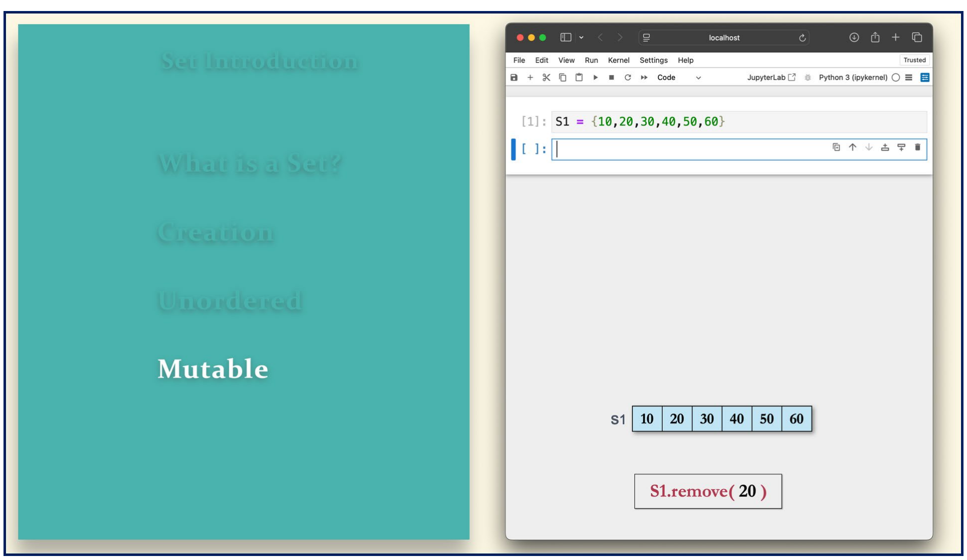
[ ]:
S1 10 20 30 40 50 60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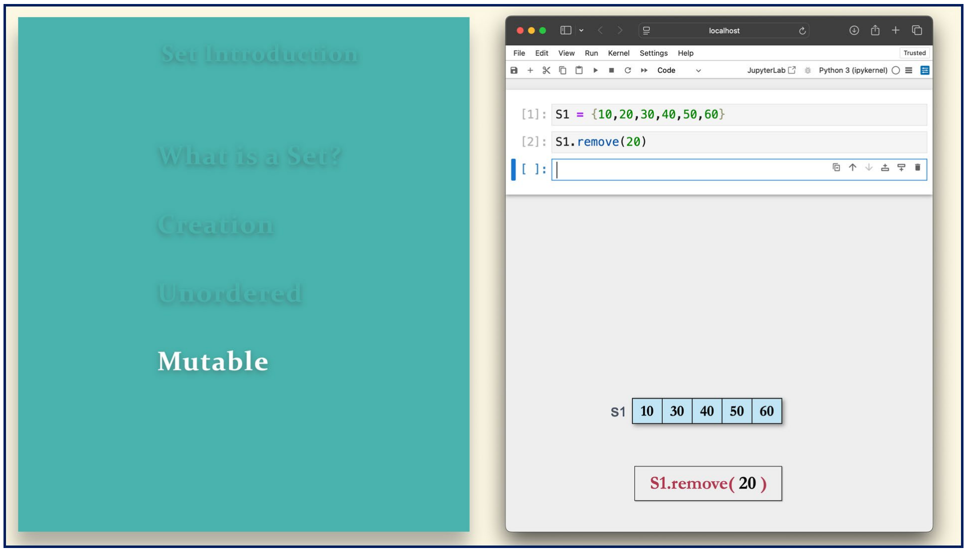
Python 3 (ipykernel)
[1]: S1 = {10,20,30,40,50,60}
[ ]:
S1 10 20 30 40 50 60
S1.remove(20)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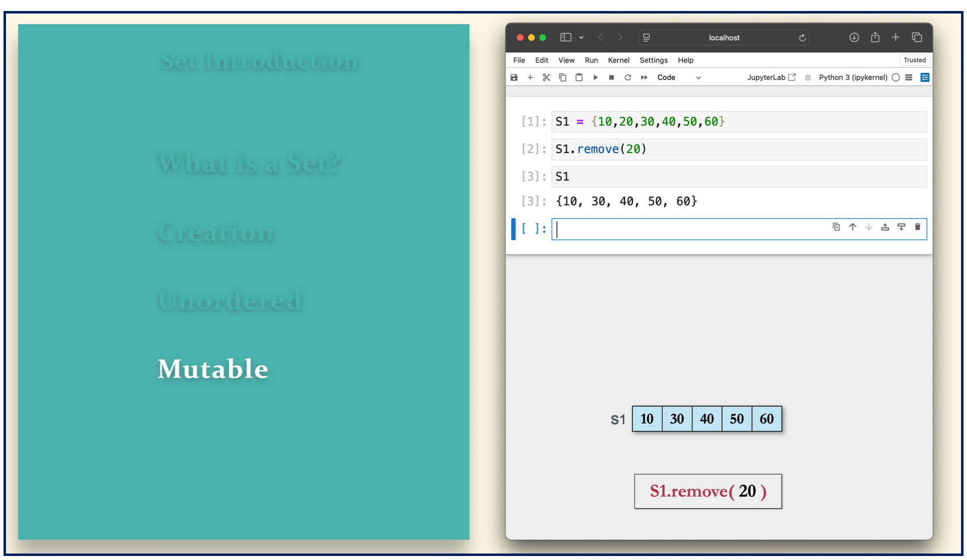
[1]: S1 = {10,20,30,40,50,60}
[2]: S1.remove(20)
[ ]:
S1
10
30
40
50
60
S1.remove(20)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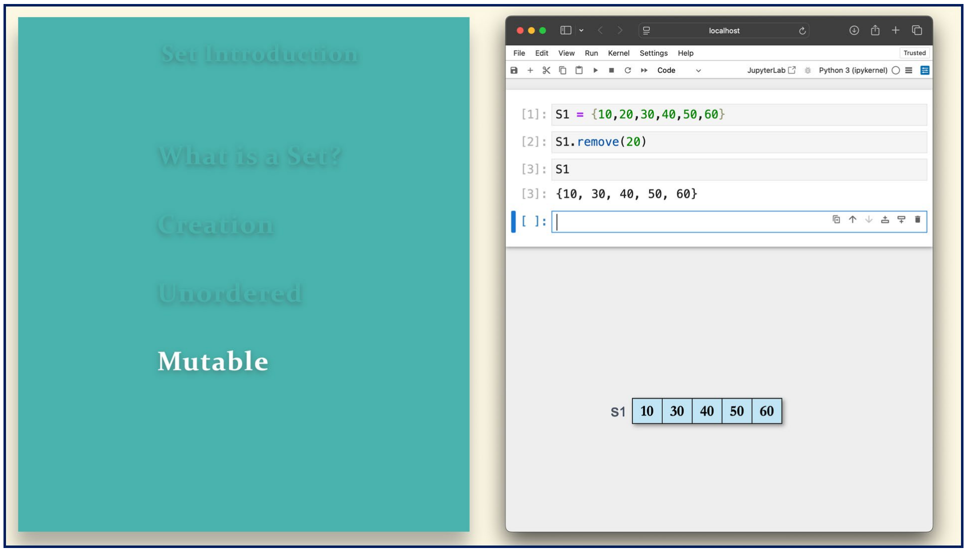
[2]: S1.remove(20)
[3]: S1
[3]: {10, 30, 40, 50, 60}
[ ]:
S1
10
30
40
50
60
S1.remove(20)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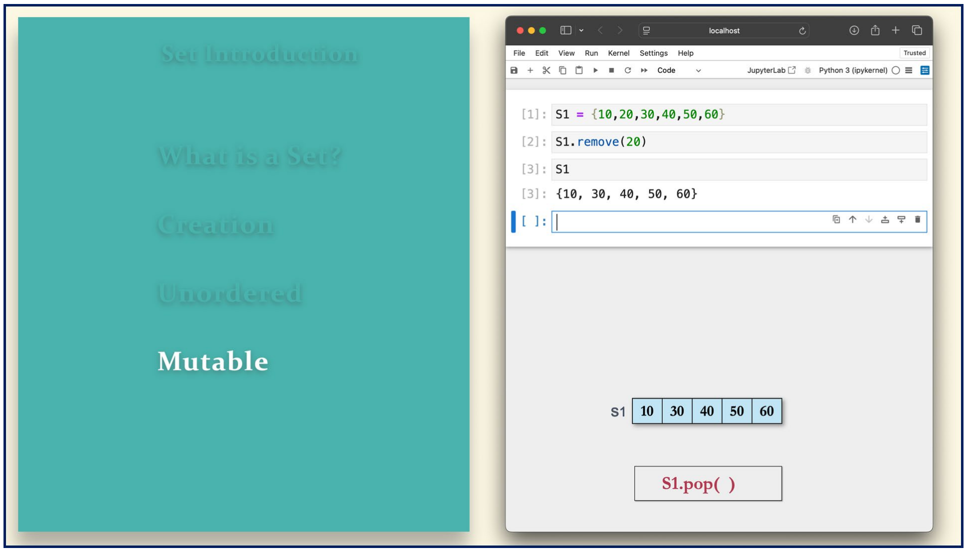
[3]: {10, 30, 40, 50, 60}
S1 10 30 40 50 60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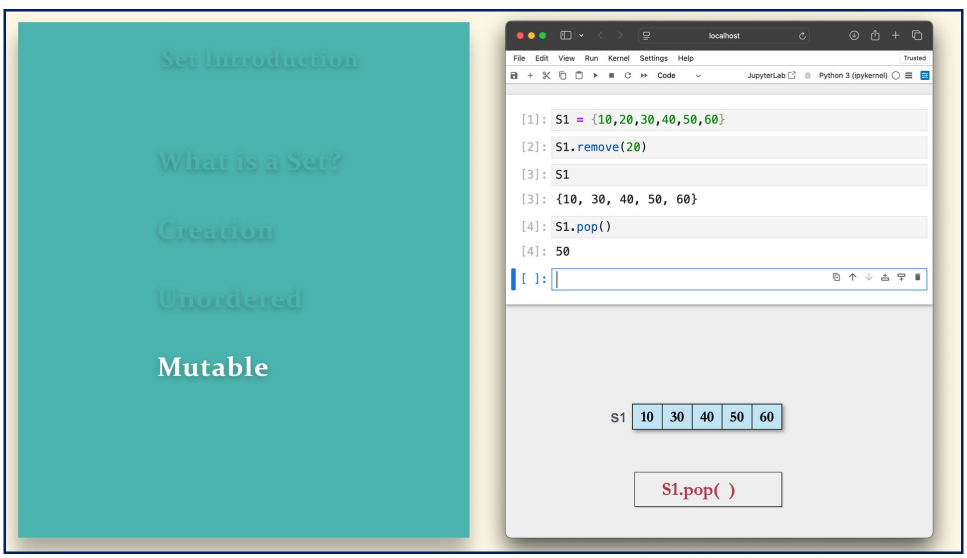
[1]: S1 = {10,20,30,40,50,60}
[2]: S1.remove(20)
[3]: S1
[3]: {10, 30, 40, 50, 60}
[ ]:
S1
10
30
40
50
60
S1.pop( )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[1]: S1 = {10,20,30,40,50,60}
[2]: S1.remove(20)
[3]: S1
[3]: {10, 30, 40, 50, 60}
[4]: S1.pop()
[4]: 50
[ ]:
S1
10
30
40
50
60
S1.pop( )